

FYS 5419/9419
Lecture
April 23,
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BASICS of SVMs

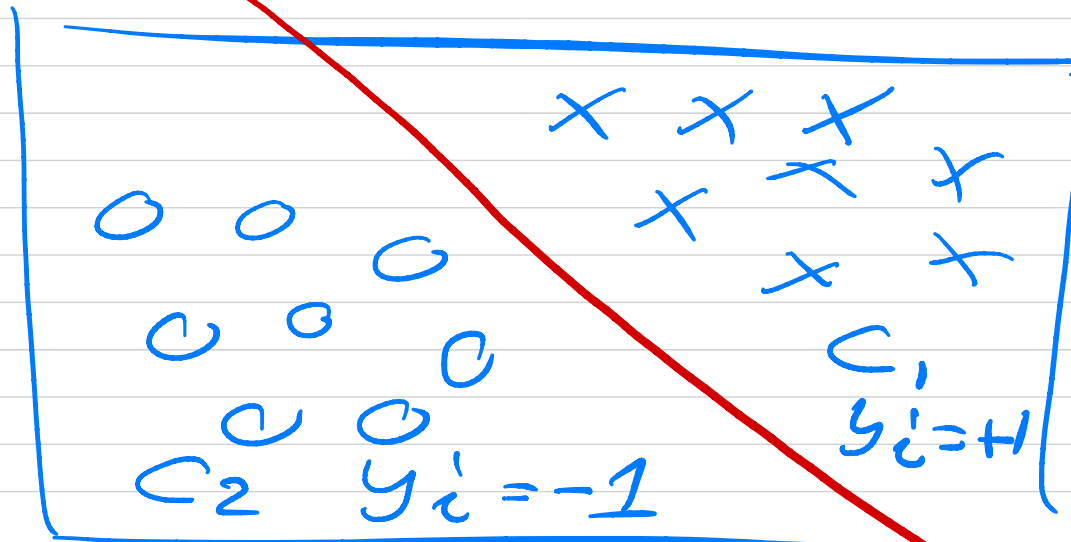
Date set with two features

$$x_1^T = [x_1(0) \quad x_1(1) \quad \dots \quad x_1(n-1)]$$

$$x_2^T = [x_2(0) \quad x_2(1) \quad \dots \quad x_2(n-1)]$$

outputs $y^T = [y_0 \quad y_1 \quad \dots \quad y_{n-1}]$

x_2



classification
problem

$$y = 0$$

x_1

$$y_i \in \{-1, 1\}$$

$$x^T = [x_1, x_2]$$

w_0 = bias ("intercept")

Model

$$\hat{y}(x) = w^T x + w_0$$

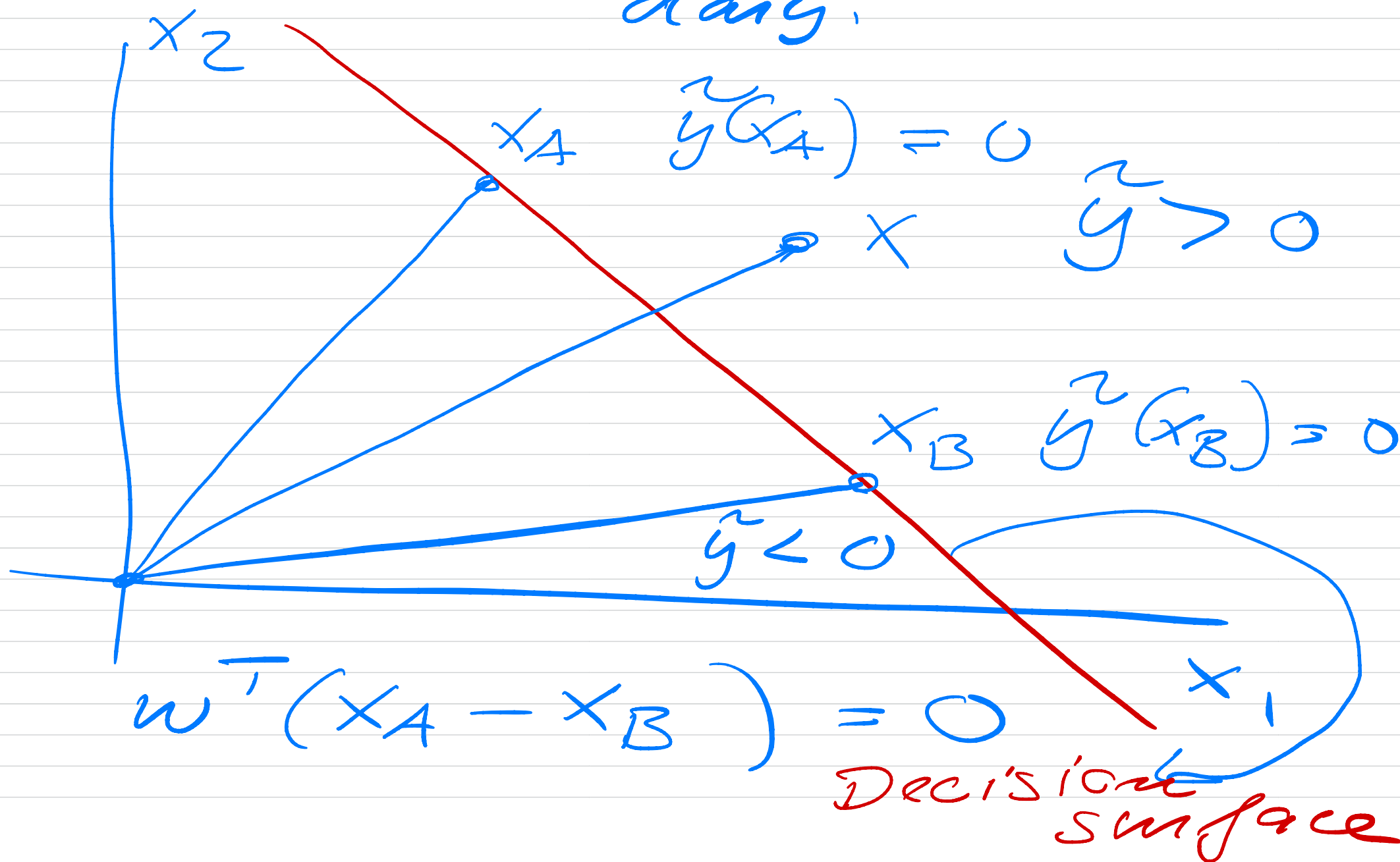
$$= w_1 x_1 + w_2 x_2 + w_0$$

$\hat{y}(x) > 0$, then the output belongs to C_1 ($y_i = +1$)

else < 0 , then it belongs to C_2

$$\tilde{y}(x) = 0$$

Defines the boundary.



$$w^T = [w_1, w_2]$$

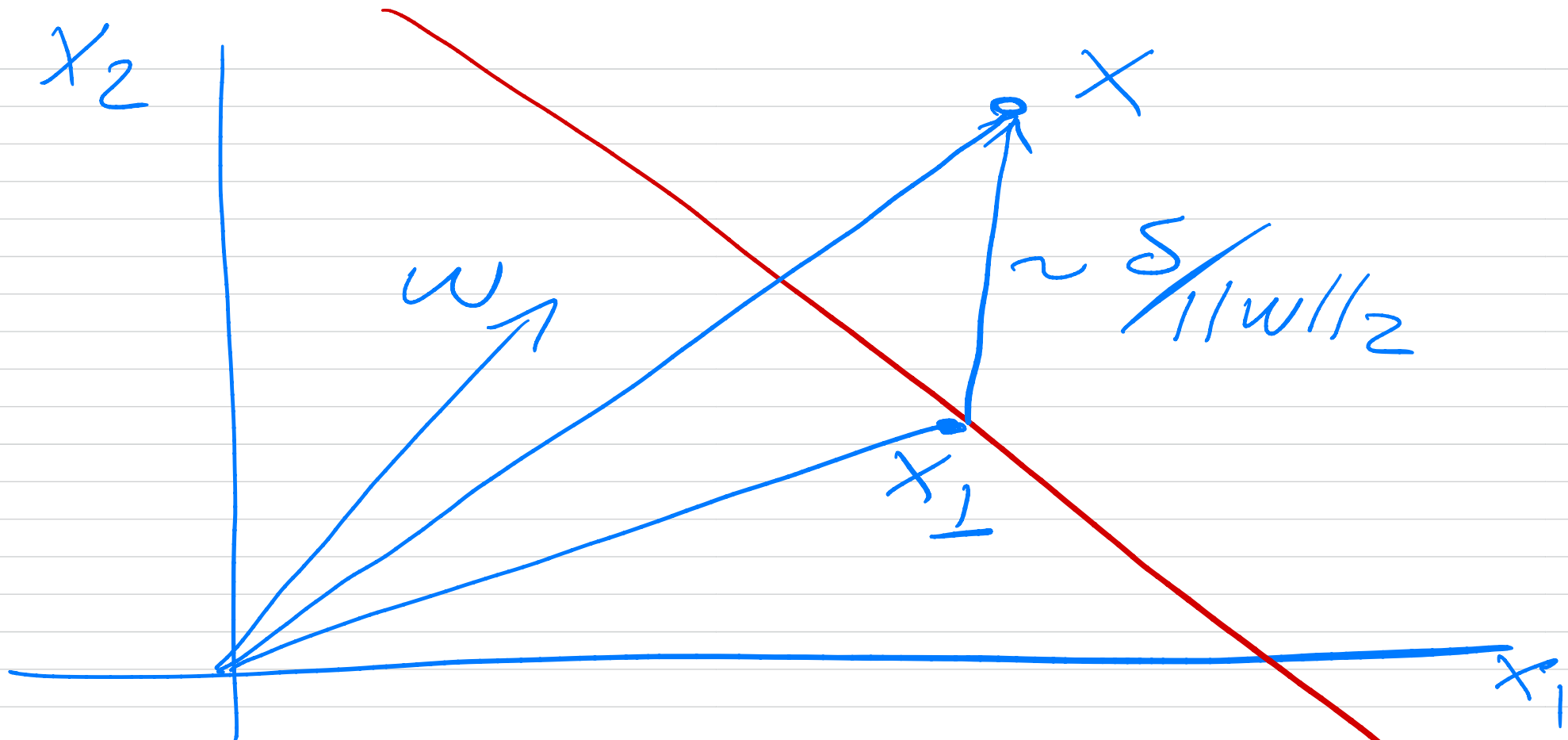
w determines the orientation of the decision surface (line) from a chosen reference

$$\tilde{y}(x) = 0 = w^T x + w_0$$

$$\tilde{y}(x_{\perp}) = 0$$

normalize w

$$\sqrt{w^T w} = \|w\|_2$$



$$w^T x = ?$$

$$\frac{w^T x}{\|w\|_2} = - \frac{w_0}{\|w\|_2}$$

$$g^2(x_{\perp}) = 0$$

$$x = \underline{x_1} + \frac{\delta w^T w}{\|w\|_2}$$

$$\delta = \frac{\tilde{y}(x)}{\|w\|_2}$$

$$y(\underline{x_1}) = w^T \underline{x_1} + w_0 = 0$$

Find w_0, w_1, w_2 by a simple optimization problem.

Define a cost/cost function
which contains all mis-
classifications (Set M)
and we want to mini-
mize it.

$$C(w, w_0) = - \sum_{i \in M} y_i (w^T x_i + w_0)$$

can be
a vector

if $y_i = 1$, misclassification
if $w^T x_i + w_0 < 0$

opposite if $y_i' = 1$

$$C(w, w_0) = - \sum_{i \in M} y_i' \tilde{y}_i$$

$$\frac{\partial C}{\partial w_0} = 0 = - \sum_{i \in M} y_i'$$

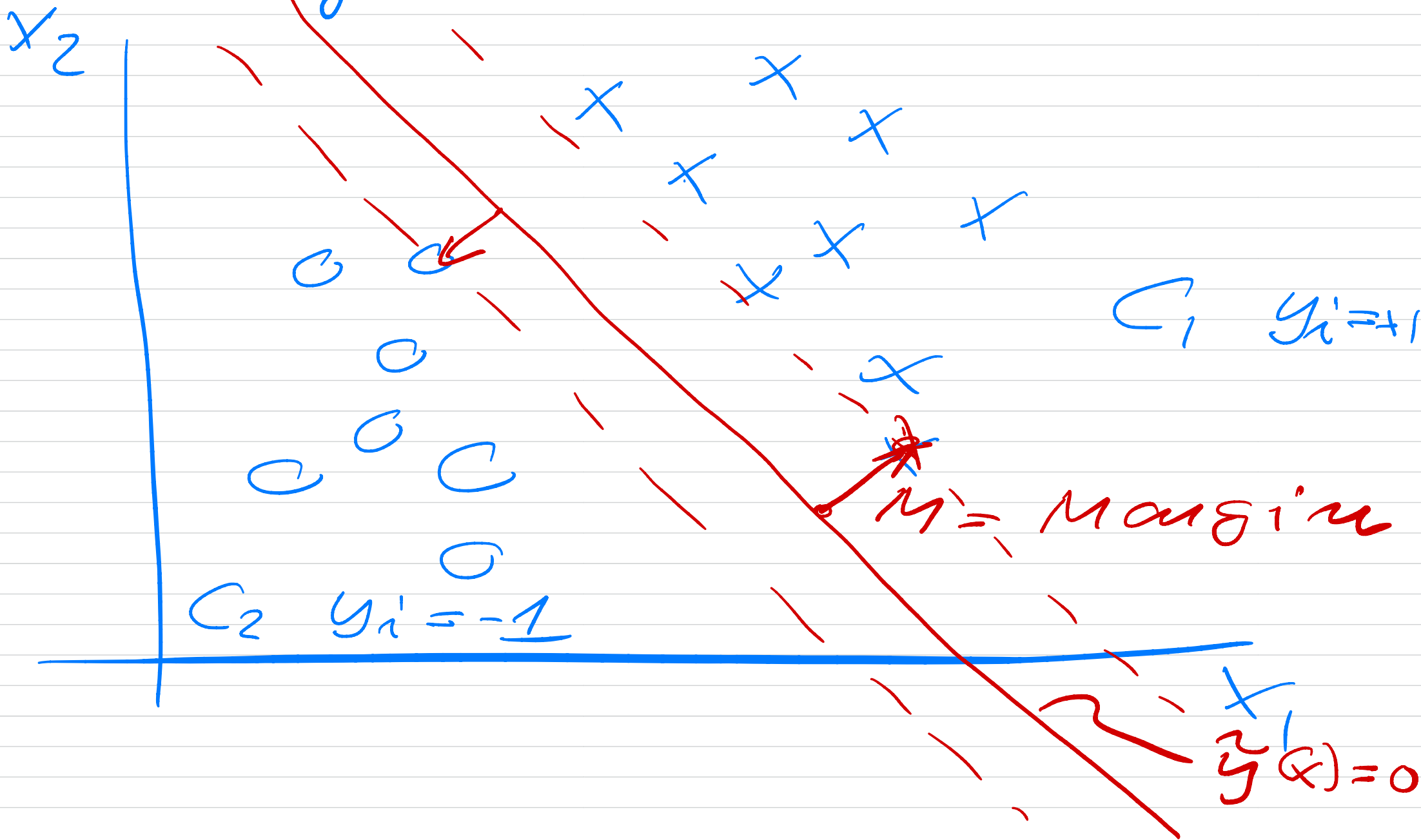
$$\frac{\partial C}{\partial w} = 0 = - \sum_{i' \in M} y_i' x_{i'}$$

update with gradient
descent

$$w_0 \leftarrow w_0 - \eta \frac{\partial C}{\partial w_0}$$

$$w \leftarrow w - \eta \frac{\partial C}{\partial w}$$

Margine: M



we seek a margin

$$y_i' \left[\frac{1}{\|w\|_2} (w^T x_i' + w_0) \right] \geq M$$

$\hat{y}_i'(x_i')$

$$y_i' \hat{y}_i' \geq 0$$

$$y_i' (w^T x_i' + w_0) \geq M \|w\|_2$$

$$M \|w\|_2 = 1 \Rightarrow M = \frac{1}{\|w\|_2}$$

Problem

$$\min \|w\|_2$$
$$w, w_0$$

with the constraint

$$y_i (w^T x_i + w_0) \geq \underline{1}$$

Example 1

$$f(x_1, x_2) = x_1^2 - x_2^2$$

$$x_1 \geq 0 \quad \wedge \quad x_2 \geq 0$$

$$\downarrow$$
$$g_1(x) = b_1$$

$$\searrow g_2(x) = b_2$$

$$\dots g_m(x) = b_m$$

$$\begin{aligned} \mathcal{L}(x, \lambda) = & f(x) - \lambda_1 (g_1(x) - b_1) \\ & - \lambda_2 (g_2(x) - b_2) - \dots \\ & \dots - \lambda_m (g_m(x) - b_m) \end{aligned}$$

$\lambda_1, \lambda_2, \dots, \lambda_m$ are Lagrange multipliers

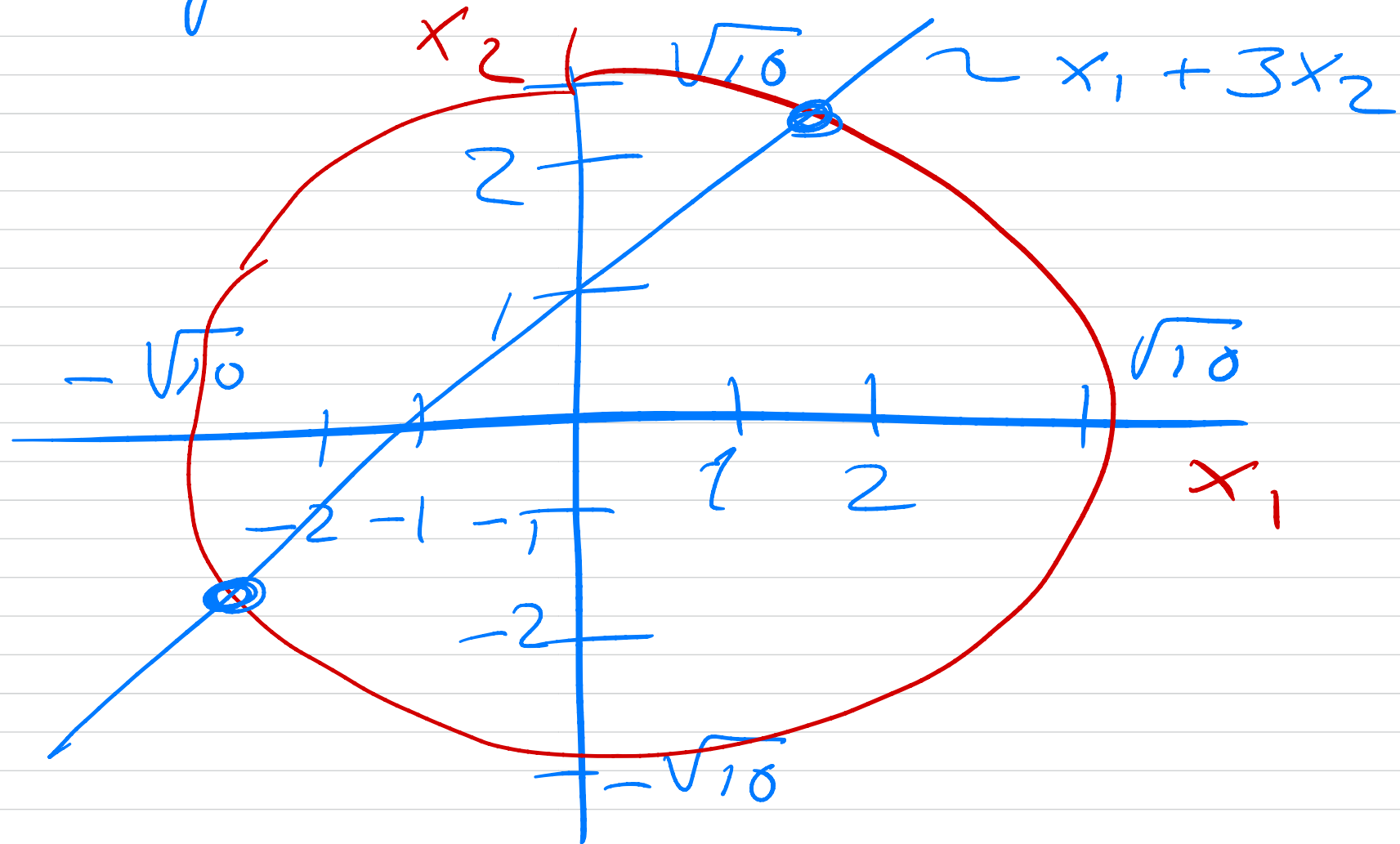
$$\frac{\partial \mathcal{L}}{\partial x_i} = 0 \quad \text{for } i = 0, 1, \dots, n-1$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_j} = 0 \quad \text{for } j = 1, 2, \dots, m$$

Example 2

$$f(x_1, x_2) = x_1 + 3x_2$$

subject to $x_1^2 + x_2^2 = 10$



$$\frac{\partial \mathcal{L}}{\partial x_1} = 0 = 1 - 2\lambda \Rightarrow x_1 = \lambda/2$$

$$\mathcal{L} = x_1 + 3x_2 - \lambda(x_1^2 + x_2^2 - 10)$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = 0 \Rightarrow x_2 = 3/2 \lambda$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 0 \Rightarrow x_1^2 + x_2^2 = 10$$

$$\Rightarrow \lambda = \pm 1/2$$

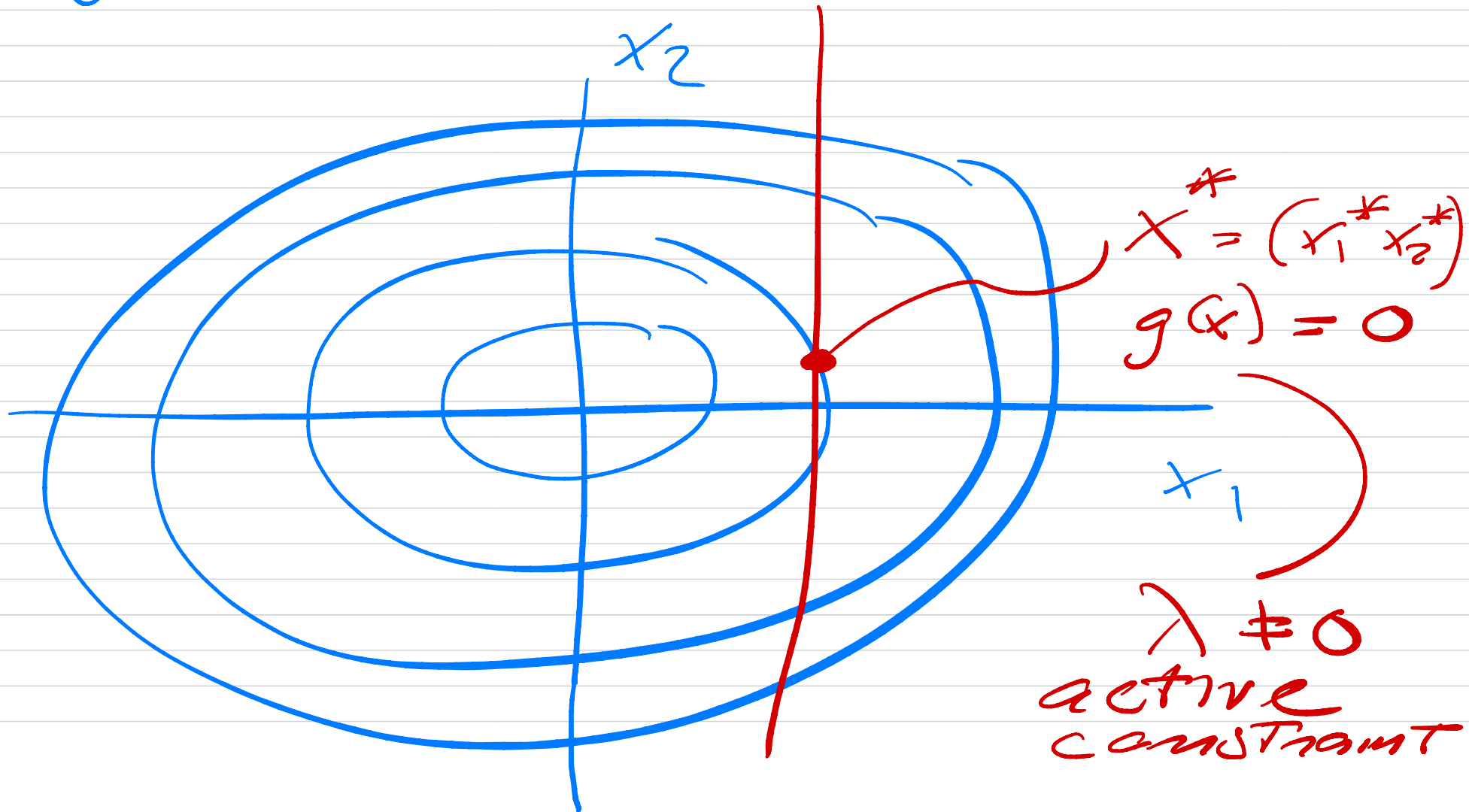
$$(i) \quad \lambda = 1/2 \quad \wedge x = \underline{1} \quad \wedge x_2 = 3$$

$$(ii) \quad \lambda = -1/2 \quad \wedge x = \underline{-1} \quad \wedge x_2 = -3$$

Example 3

$$f(x_1, x_2) = 1 - x_1^2 - x_2^2 \quad \text{subject to}$$

$$g(x_1, x_2) = x_1 + x_2 - 1$$



inactive $g(x) > 0$
 $\nabla_x f(x) = 0$ and have $\lambda = 0$

For either case $\lambda \cdot g(x) = 0$
we optimize L wrt. x
and λ subject to

$$\begin{array}{l} g(x) \geq 0 \quad \lambda \geq 0 \\ \lambda \cdot g(x) = 0 \end{array}$$

SUM

$$\min_{w, w_0} \frac{1}{2} w^T w$$

subject to

$$y_i (w^T x_i + w_0) = 1 \quad \forall i=0, 1, \dots, n-1$$

$$\mathcal{L} = \frac{1}{2} w^T w - \sum_{i=0}^{n-1} \lambda_i [y_i (w^T x_i + w_0) - 1]$$

$$\frac{\partial \mathcal{L}}{\partial w_0} = 0 = - \sum y_i \lambda_i$$

$$\frac{\partial \mathcal{L}}{\partial w^T} = 0 = w - \sum_{i=0}^{n-1} \lambda_i' g_i' x_i' \quad \uparrow \text{vector}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_i'} = 0 = g_i' (w^T x_i' + w_0) - 1$$

$$w = \sum_i \lambda_i' g_i' x_i', \text{ insert in } \mathcal{L}$$

$$\mathcal{L} = \sum_{i=0}^{n-1} \lambda_i' - \frac{1}{2} \sum_{i,j} \lambda_i' \lambda_j' g_i' g_j' x_i^T x_j$$

subject to $\lambda_i' \geq 0 \wedge \sum \lambda_i' g_i' = 0$
 $\wedge [g_i' (w^T x_i' + w_0) - 1]$

$$g_i (w^T x_i' + w_0) = 1$$

$$w_0 = \frac{1}{g_i} - w^T x_i'$$

$$\overline{X} = [\lambda_0 \lambda_1 \dots \lambda_{n-1}]$$

$$\overline{y} = [y_0 y_1 \dots y_{n-1}]$$

$$\mathcal{L} = \frac{1}{2} \lambda^T \begin{bmatrix} y_0 y_0 x_0^T x_0 & y_0 y_1 x_0^T x_1 & \dots & y_0 y_{n-1} x_0^T x_{n-1} \\ \vdots & & & \\ y_{n-1} y_0 x_{n-1}^T x_0 & \dots & \dots & y_{n-1} y_{n-1} x_{n-1}^T x_{n-1} \end{bmatrix} \lambda -$$

$$\times \lambda$$

subject to $y^T \lambda = 0$

$$\lambda_i \geq 0 \quad y_i' (w^T x_i' + w_0) - 1 = 0$$

Replace $x_0^T x_0, \dots, x_n^T x_n$
with a so-called kernel

$$K(x_i, x_j)$$

$$P = \begin{bmatrix} y_0 y_0 K(x_0, x_0) & \dots & y_0 y_{n-1} K(x_0, x_{n-1}) \\ \vdots & \ddots & \vdots \\ y_{n-1} y_{n-1} K(x_{n-1}, x_{n-1}) \end{bmatrix}$$

$$L(\lambda) = \mathbb{1}\lambda - \frac{1}{2} \lambda^T \Phi \lambda$$

$$K(x_i, x_j) = (\phi(x_i))^T \phi(x_j)$$

linear kernel $\phi(x_i) = x_i$